

CPSC 304 Project Cover Page

Milestone #: 4

Date: Nov 26, 2021

Group Number: 91

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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above. (In the case of Project Milestone 0, the main purpose of this page is for you to let us know your email address, and then let us assign you to a TA for your project supervisor.)

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

2. Single SQL script: Handed in on Canvas

3. A. Description of the final project, and what it accomplished.

Our project is a cookbook application that contains recipes for dishes.

Our members can login with their User ID. If the user is not our member, he or she can sign up easily.

Our member can search recipe by:

- (1) recipe name
- (2) ingredient name
- (3) cooking method

Members can see the average rate of a recipe.

If member does not know what to search, they can get recommended recipes by :

- (1) the highest rated recipe
- (2) the number of comment for each recipe name
- (3) the number of comments(>2) for each recipe name

Members can get recipes by filtering milk or fat or bacon.

Manager can delete the recipes.

B. How our final schema differed from the schema we turned in and why:

(1) Change: We added name and detail as attribute to entity recipe. Deleting user_id attribute.
Reason: user_id is a foreign key so delete. Recipe need recipe name and cooking step.

(2) Change: Changing the relation between entity recipe and entity typeByMethod from all recipes having only one typeByMethod to all recipes having at least one typeByMethod. (recipe and typeByMethod was many-to-one, now is many-to-many)
Reason: One recipe may need multiple cooking methods.

(3) Change: Adding photo_url as attribute to entity photos.
Reason: Each photo needs a URL.

(4) Change: Changing the relation between entity user account and entity location, from all user have at least one location to all user have only one location. (user_account and location was many-to-many, now is one-to-many)
Reason: That's enough for users to have one address.

(5) Change: Adding a postal code, house number and street as attribute to location.
Reason: Location should have postal code, house number and street as attribute to location.

(6) Change: Changing comment from as a relation to as an entity. The relationship between comment and user account is that all comments have and have only one user. The relationship between comment and recipe is that all comments have and have only one recipe. (comment

and recipe was many-to-many, now is many-to-one. comment and user_account was many-to-many, now is many-to-one)

Reason: All comments should be written by the user and belonging to a recipe. A user can write multiple comments to a recipe, so change it to entity.

(7) Change: Changing rating from as an entity to as a relation.

Reason: Members rate a recipe once.

C&D: Screenshots of the sample output of queries using the GUI (code reference - data table before run query- GUI input - data table after run query)

There is the url which can direct to our webpage:

<https://www.students.cs.ubc.ca/~jeanoo/loginX.php>

<https://www.students.cs.ubc.ca/~jeanoo/userInterface.php> (Secondary site, after logged in as a user account)

<https://www.students.cs.ubc.ca/~jeanoo/managerInterface.php> (Secondary site after logged in as a manager account)

- **1. Insertion:**

@loginX.php

Code reference:

```
loginX.php
271
272 function handleSignupRequest(){
273     global $db_conn;
274     $MAX1=executePlainSQL("SELECT ACCOUNT_ID FROM account WHERE account_id >= all(SELECT account_id FROM account)");
275     $row = OCI_Fetch_Array($MAX1, OCI_BOTH);
276     //echo "<tr><td>" . $row["ACCOUNT_ID"] . "</td><td>" . $row["USERNAME"] . "</td></tr>";
277     //echo $row[0];
278     $ID=(int)$row[0]+1;
279
280     echo "<tr><td>Welcome, your new ID is " . $ID . "</td></tr>";
281
282     //insert into ACCOUNT
283     $userName=$_POST['userName'];
284     $email=$_POST['email'];
285     $PSWD=$_POST['PSWD'];
286
287     $tuple = array (
288         ":bind1" => $ID,
289         ":bind2" => $_POST['userName'],
290         ":bind3" => $_POST['email'],
291         ":bind4" => $_POST['PSWD'],
292     );
293
294     $alltuples = array (
295         $tuple
296     );
297
298     executeBoundSQL("insert into ACCOUNT values (:bind1, :bind2, :bind3, :bind4)", $alltuples);
299
```

Before insertion:

ACCOUNT				
10 rows				
Tx: Manual				
DDL				
WHERE				
ORDER BY				
	ACCOUNT_ID	USERNAME	EMAIL	ACCOUNT_PASSWORD
1	1	Cindy Thorne	1@gmail.com	1
2	2	Zander Villalobos	2@gmail.com	2
3	3	Pooja Croft	3@gmail.com	3
4	4	Alma Drew	4@gmail.com	4
5	5	Loren Bowen	5@gmail.com	5
6	6	Alessia Mayer	6@gmail.com	6
7	7	Helena Cuevas	7@gmail.com	7
8	8	Gregor Hope	8@gmail.com	8
9	9	Alfie Dean	9@gmail.com	9
10	10	Jarrad Hatfield	10@gmail.com	10

ACCOUNT

USER_ACCOUNT

<

>

9 rows

<

>

↺

+

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Tx: Manual

✓

↺

DDL

🔍

CSV

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⌄

↺

USER_ACCOUNT

▼

WHERE

ORDER BY

	USER_ACCOUNT_ID	COOKING_LEVEL	POSTAL_CODE	HOUSE#	STREET
1	2	3	d8336n	2222 589	Darwin Ln.
2	3	5	438	1111 67	Seventh Av.
3	4	3	d7fh2b	4444 3	Balding Pl.
4	5	1	38i	5555 1956	Arlington Pl.
5	6	2	d83m	6666 301	Putnam
6	7	2	d3g	7777 5420	Telegraph Av.
7	8	4	ddd	8888 5420	College Av.
8	9	3	d9j3	9999 5720	McAuley St.
9	10	5	dk3	0 44	Upland Hts.

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ACCOUNT	USER_ACCOUNT	LOCATION1
10 rows		
WHERE	ORDER BY	
POSTAL_CODE	HOUSE#	STREET
1 38i	5555	1956 Arlington Pl.
2 438	1111	67 Seventh Av.
3 V6R 2M6	4643	11Th Ave W
4 d3g	7777	5420 Telegraph Av.
5 d7fh2b	4444	3 Balding Pl.
6 d8336n	2222	589 Darwin Ln.
7 d83m	6666	301 Putnam
8 d9j3	9999	5720 McAuley St.
9 ddd	8888	5420 College Av.
10 dk3	0	44 Upland Hts.

ACCOUNT	USER_ACCOUNT	LOCATION1	LOCATION2
11 rows			
WHERE	ORDER BY		
POSTAL_CODE	CITY	PROVINCE	
1 v7gd3n	Vancouver	BC	
2 d8336n	Edmonton	Alberta	
3 438	Winnipeg	Manitoba	
4 d7fh2b	Victoria	BC	
5 38i	Fredericton	NB	
6 d83m	St.Johns	Newfoundland	
7 d3g	Halifax	NS	
8 ddd	Toronto	Ontario	
9 d9j3	Charlottetown	PEI	
10 dk3	QuebecCity	Quebec	
11 V6R 2M6	Vancouver	BC	

GUI input – insertion

The image shows a 'Sign Up Today' form with a light brown background and a darker brown border. The form contains the following fields and labels:

- UserName:** A text input field containing 'Alice'.
- Email:** A text input field containing 'Alice@gmail.com'.
- Password:** A text input field containing '123'.
- Cooking Level:** A text input field containing '2'.
- Postal Code:** A text input field containing 'v6r3r3'.
- City:** A text input field containing 'Vancouver'.
- Province:** A text input field containing 'British Columbia'.
- House Number:** A text input field containing '5787'.
- Street:** A text input field containing 'cypress st.'.
- Sign Up:** A button at the bottom of the form.

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After insertion:

ACCOUNT	USER_ACCOUNT	LOCATION1	LOCATION2
<	<	11 rows	>
+	-	↶	↷
Tx: Manual	✓	DDL	Q
CSV	↓	↑	→
WHERE	ORDER BY		
ACCOUNT_ID	USERNAME	EMAIL	ACCOUNT_PASSWORD
1	Cindy Thorne	1@gmail.com	1
2	Zander Villalobos	2@gmail.com	2
3	Pooja Croft	3@gmail.com	3
4	Alma Drew	4@gmail.com	4
5	Loren Bowen	5@gmail.com	5
6	Alessia Mayer	6@gmail.com	6
7	Helena Cuevas	7@gmail.com	7
8	Gregor Hope	8@gmail.com	8
9	Alfie Dean	9@gmail.com	9
10	Jarrad Hatfield	10@gmail.com	10
11	Alice	Alice@gmail.com	123

ACCOUNT

USER_ACCOUNT

LOCATION1

LOCATION2

<

>

10 rows

>

>|

↺

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+

-

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↶

↷

Tx: Manual

✓

↺

DDL

Q

CSV

↓

↑

↺

↻

USER_ACCOUNT

WHERE

ORDER BY

	USER_ACCOUNT_ID	COOKING_LEVEL	POSTAL_CODE	HOUSE#	STREET
1	2	3	d8336n	2222	589 Darwin Ln.
2	3	5	438	1111	67 Seventh Av.
3	4	3	d7fh2b	4444	3 Balding Pl.
4	5	1	38i	5555	1956 Arlington Pl.
5	6	2	d83m	6666	301 Putnam
6	7	2	d3g	7777	5420 Telegraph Av.
7	8	4	ddd	8888	5420 College Av.
8	9	3	d9j3	9999	5720 McAuley St.
9	10	5	dk3	0	44 Upland Hts.
10	11	2	v6r3r3	5787	cypress st.

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	POSTAL_CODE	HOUSE#	STREET
1	38i	5555	1956 Arlington Pl.
2	438	1111	67 Seventh Av.
3	V6R 2M6	4643	11Th Ave W
4	d3g	7777	5420 Telegraph Av.
5	d7fh2b	4444	3 Balding Pl.
6	d8336n	2222	589 Darwin Ln.
7	d83m	6666	301 Putnam
8	d9j3	9999	5720 McAuley St.
9	ddd	8888	5420 College Av.
10	dk3	0	44 Upland Hts.
11	v6r3r3	5787	cypress st.

	POSTAL_CODE	CITY	PROVINCE
1	v7gd3n	Vancouver	BC
2	d8336n	Edmonton	Alberta
3	438	Winnipeg	Manitoba
4	d7fh2b	Victoria	BC
5	38i	Fredericton	NB
6	d83m	St.Johns	Newfoundland
7	d3g	Halifax	NS
8	ddd	Toronto	Ontario
9	d9j3	Charlottetown	PEI
10	dk3	QuebecCity	Quebec
11	V6R 2M6	Vancouver	BC
12	v6r3r3	Vancouver	British Columbia

● 2. Deletion Operation

@managerInterface.php

Code reference:

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```
managerInterface.php x # style.css # style_m.css background1.png w
managerInterface.php
145
146 //delete recipe by given recipe id
147 function handleDeletionRequest(){
148     global $db_conn;
149     $RECIPEID = $_POST['deleteID'];
150     $numRecipe = executePlainSQL("SELECT COUNT(*) FROM RECIPE_HAS");
151     if (is_numeric($RECIPEID) && $RECIPEID > 0) {
152         echo 'Delete ', $RECIPEID, 'th recipe.';
153         executePlainSQL("DELETE FROM RECIPE_HAS WHERE ID = '". $RECIPEID . "'");
154         echo "Delete Successful! <br>";
155         OCICommit($db_conn);
156     } else echo 'invalid input';
157 }
```

Database(before):

USER_ACCOUNT x ACCOUNT x RECIPE_HAS x							
Tx: Manual ✓ ↺ DDL 🔍 CSV ⬇ ⬆ ⬇ ⬇ ⬇ RECIPE_HAS ⚙							
WHERE				ORDER BY			
		CREATETIME	USER_ACCOUNT_ID	REGION_ID	RECIPE_NAME	DETAILS	DIFFICULTY
1	3	2020-06-17		3	1 Kung Pao Chicken	r3: first step...	3
2	4	2020-06-18		4	4 Cal Smart Steak	r4: first step...	4
3	5	2020-06-19		5	5 Beef Udon	r5: first step...	5
4	6	2020-06-20		6	6 Poutine	r6: first step...	4
5	7	2020-06-21		7	7 Slow Cooker Pot Roast	r7: first step...	3
6	8	2020-06-22		8	8 Pasta alla Norma	r8: first step...	2
7	9	2020-06-23		9	9 Okonomiyaki	r9: first step...	1
8	10	2020-06-23		10	10 Barbecued Pork	r10: first step...	2
9	11	2020-06-16		10	3 Taiwanese-Style Three Cup Chicken	r11: first step...	2

GUI input:

Delete a recipe

Recipe ID you want to delete:

Delete

Retrieved data from table:

ID	RECIPE_NAME	PICTURE
3	Kung Pao Chicken	
4	Cal Smart Steak	
5	Beef Udon	
6	Poutine	

database(after):

Delete a recipe

Recipe ID you want to delete:

Retrieved data from table:

ID	RECIPE_NAME	PICTURE
4	Cal Smart Steak	
5	Beef Udon	
6	Poutine	

We can see the recipe with id3 is deleted.

USER_ACCOUNT × ACCOUNT × RECIPE_HAS ×							
8 rows							
WHERE ORDER BY							
		CREATETIME	USER_ACCOUNT_ID	REGION_ID	RECIPE_NAME	DETAILS	DIFFICULTY
1	4	2020-06-18		4	4 Cal Smart Steak	r4: first step...	4
2	5	2020-06-19		5	5 Beef Udon	r5: first step...	5
3	6	2020-06-20		6	6 Poutine	r6: first step...	4
4	7	2020-06-21		7	7 Slow Cooker Pot Roast	r7: first step...	3
5	8	2020-06-22		8	8 Pasta alla Norma	r8: first step...	2
6	9	2020-06-23		9	9 Okonomiyaki	r9: first step...	1
7	10	2020-06-23		10	10 Barbecued Pork	r10: first step...	2
8	11	2020-06-16		10	3 Taiwanese-Style Three Cup Chicken	r11: first step...	2

Since the foreign key of photos_includes table is *deleted on cascade*, the photo which references the deleted photo was deleted together.

USER_ACCOUNT × ACCOUNT × RECIPE_HAS × PHOTOS_INCLUDES ×				
8 rows				
WHERE ORDER BY				
	PHOTO_ID	RECIPE_ID	PHOTO_URL	
1	4	4	/~jeanoo/4.png	
2	5	5	/~jeanoo/5.png	
3	6	6	/~jeanoo/6.png	
4	7	7	/~jeanoo/7.png	
5	8	8	/~jeanoo/8.png	
6	9	9	/~jeanoo/9.png	
7	10	10	/~jeanoo/10.png	
8	11	11	/~jeanoo/1.png	

● 3. Update Operation

code reference:

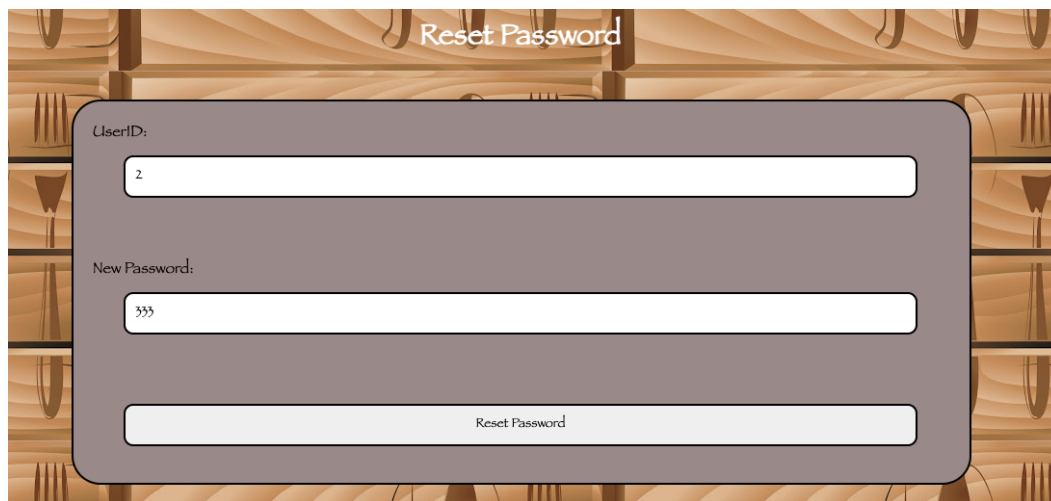
In loginX.php

```
loginX.php × # style_m.css managerInterface.php userInterface.php background1.png # style.css
loginX.php
201 function handleResetPasswordRequest() {
202     global $db_conn;
203
204     $RUserID = $_POST['RUserID'];
205     $newPSWD = $_POST['newPSWD'];
206
207     // you need the wrap the old name and new name values with single quotations
208     executePlainSQL("UPDATE ACCOUNT SET ACCOUNT_PASSWORD='". $newPSWD . "' WHERE ACCOUNT_ID='". $RUserID . "'");
209     echo "<br> Reset Successfully! <br>";
210     // $RIGHT_PASSWORD = executePlainSQL("SELECT ACCOUNT_PASSWORD FROM ACCOUNT WHERE ACCOUNT_ID='". $RUserID . "'");
211     // $row = OCI_Fetch_Array($RIGHT_PASSWORD, OCI_BOTH);
212     // echo "<tr><td>" . "newpass".$row["ACCOUNT_PASSWORD"] . "</td></tr>";
213
214     OCICommit($db_conn);
215 }
```

database(before):

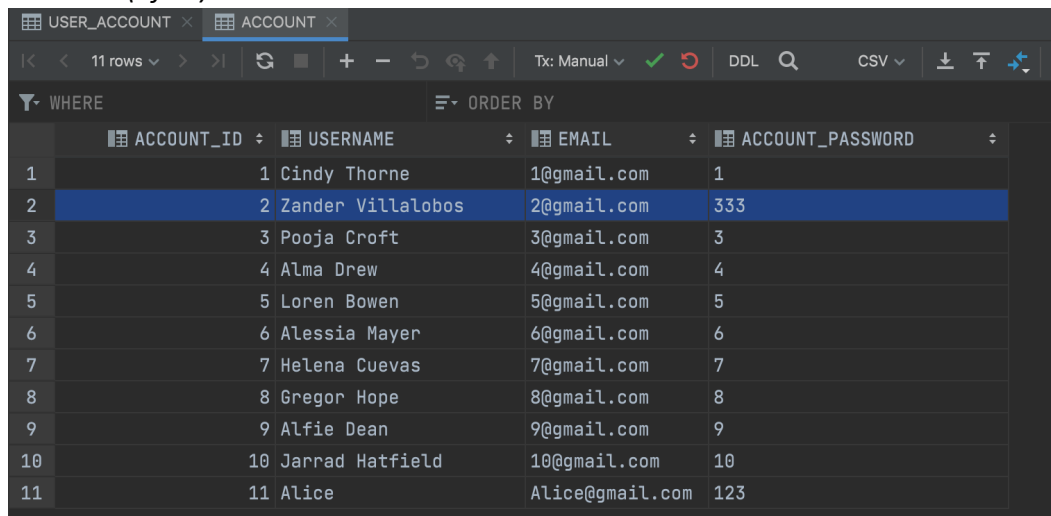
USER_ACCOUNT ×		ACCOUNT ×	
10 rows		Tx: Manual	
WHERE		ORDER BY	
	ACCOUNT_ID	USERNAME	EMAIL
1	1	Cindy Thorne	1@gmail.com
2	2	Zander Villalobos	2@gmail.com
3	3	Pooja Croft	3@gmail.com
4	4	Alma Drew	4@gmail.com
5	5	Loren Bowen	5@gmail.com
6	6	Alessia Mayer	6@gmail.com
7	7	Helena Cuevas	7@gmail.com
8	8	Gregor Hope	8@gmail.com
9	9	Alfie Dean	9@gmail.com
10	10	Jarrad Hatfield	10@gmail.com

GUI input:



A web form titled "Reset Password" with a wood-grain background. The form is a light gray rounded rectangle containing three input fields and a button. The first field is labeled "UserID:" and contains the number "2". The second field is labeled "New Password:" and contains "333". The third field is a button labeled "Reset Password".

Database(after):



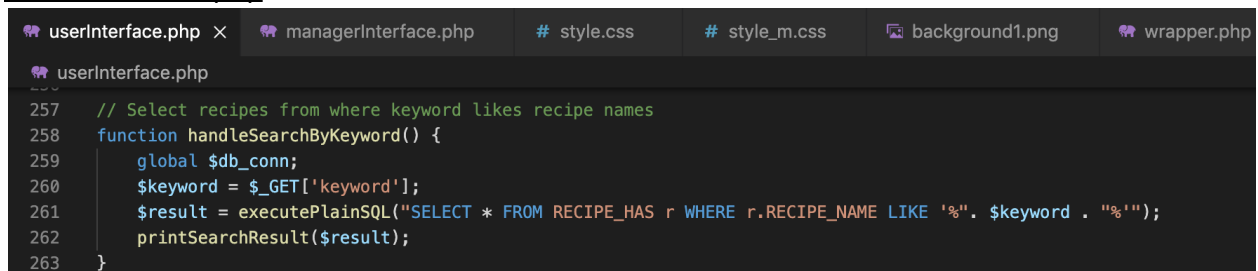
A screenshot of a database management tool showing a table named "ACCOUNT". The table has four columns: "ACCOUNT_ID", "USERNAME", "EMAIL", and "ACCOUNT_PASSWORD". There are 11 rows of data. The second row, with ID 2, is highlighted in blue and shows the username "Zander Villalobos" and password "333".

ACCOUNT_ID	USERNAME	EMAIL	ACCOUNT_PASSWORD
1	Cindy Thorne	1@gmail.com	1
2	Zander Villalobos	2@gmail.com	333
3	Pooja Croft	3@gmail.com	3
4	Alma Drew	4@gmail.com	4
5	Loren Bowen	5@gmail.com	5
6	Alessia Mayer	6@gmail.com	6
7	Helena Cuevas	7@gmail.com	7
8	Gregor Hope	8@gmail.com	8
9	Alfie Dean	9@gmail.com	9
10	Jarrad Hatfield	10@gmail.com	10
11	Alice	Alice@gmail.com	123

● 4. Selection

Code reference:

@userInterface.php



A screenshot of a code editor showing PHP code in a file named "userInterface.php". The code defines a function "handleSearchByKeyword()" that uses a database connection to search for recipes by keyword.

```
257 // Select recipes from where keyword likes recipe names
258 function handleSearchByKeyword() {
259     global $db_conn;
260     $keyword = $_GET['keyword'];
261     $result = executePlainSQL("SELECT * FROM RECIPE_HAS r WHERE r.RECIPE_NAME LIKE '%" . $keyword . "%'");
262     printSearchResult($result);
263 }
```

Database(before):

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```
CREATE TABLE recipe_has (  
    ID                INTEGER PRIMARY KEY,  
    CREATETIME        DATE,  
    USER_ACCOUNT_ID   INTEGER NOT NULL,  
    REGION_ID         INTEGER NOT NULL,  
    RECIPE_NAME        CHAR(50) NOT NULL,  
    DETAILS            CHAR(500),  
    DIFFICULTY         SMALLINT,  
    FOREIGN KEY (USER_ACCOUNT_ID) REFERENCES user_account(USER_ACCOUNT_ID) ON DELETE CASCADE,  
    FOREIGN KEY (REGION_ID) REFERENCES typeByRegion1(ID) ON DELETE CASCADE);  
grant select on recipe_has to public;
```

```
insert into recipe_has  
values(1, To_Date('16 Jun 2020','DD Mon YYYY'),4,3,'Taiwanese-Style Three Cup Chicken','r1: first step...',1);  
insert into recipe_has  
values(2,To_Date('30 Dec 2020','DD Mon YYYY'),2,2,'Sweet and Sour Pork','r2: first step...',2);  
insert into recipe_has  
values(3,To_Date('17 Jun 2020','DD Mon YYYY'),3,1,'Kung Pao Chicken','r3: first step...',3);  
insert into recipe_has  
values(4,To_Date('18 Jun 2020','DD Mon YYYY'),4,4,'Cal Smart Steak','r4: first step...',4);  
insert into recipe_has  
values(5,To_Date('19 Jun 2020','DD Mon YYYY'),5,5,'Beef Udon','r5: first step...',5);  
insert into recipe_has  
values(6,To_Date('20 Jun 2020','DD Mon YYYY'),6,6,'Poutine','r6: first step...',4);  
insert into recipe_has  
values(7,To_Date('21 Jun 2020','DD Mon YYYY'),7,7,'Slow Cooker Pot Roast','r7: first step...',3);  
insert into recipe_has  
values(8,To_Date('22 Jun 2020','DD Mon YYYY'),8,8,'Pasta alla Norma','r8: first step...',2);  
insert into recipe_has  
values(9,To_Date('23 Jun 2020','DD Mon YYYY'),9,9,'Okonomiyaki','r9: first step...',1);  
insert into recipe_has  
values(10,To_Date('23 Nov 2021','DD Mon YYYY'),10,10,'Barbecued Pork','r10: first step...',2);  
insert into recipe_has  
values(11, To_Date('16 Jun 2020','DD Mon YYYY'),10,3,'Taiwanese-Style Three Cup Chicken','r11: first step...',2);
```

GUI input:

Search

Search recipes by keywords

Keyword of Recipe Name(capital the initial letter):

Taiwanese-Style Three Cup Chicken

Search

Result:

Retrieved data from table:			
ID	RECIPE_NAME	STEPS	PICTURE
11	Taiwanese-Style Three Cup Chicken	rl1: first step...	

- 5. Projection

Code reference:

@userInterface.php

```
userInterface.php • managerInterface.php # style.css # style_m.css background1.png
userInterface.php
264 // project into r.ID and r.recipe_name
265 function handleSearchByMethod() {
266     global $db_conn;
267     $keyword = $_GET['keyword'];
268     $result = executePlainSQL("SELECT r.ID, r.RECIPE_NAME
269 FROM RECIPE_HAS r, typebymethod m, recipeMethod rm
270 WHERE m.method LIKE '%" . $keyword . "%' and m.ID = rm.method_ID and rm.recipe_ID = r.ID");
271     printSearchResult($result);
272 }
```

Database(before):

```
CREATE TABLE recipe_has (
    ID                INTEGER PRIMARY KEY,
    CREATETIME        DATE,
    USER_ACCOUNT_ID   INTEGER NOT NULL,
    REGION_ID         INTEGER NOT NULL,
    RECIPE_NAME        CHAR(50) NOT NULL,
    DETAILS            CHAR(500),
    DIFFICULTY         SMALLINT,
    FOREIGN KEY (USER_ACCOUNT_ID) REFERENCES user_account(USER_ACCOUNT_ID) ON DELETE CASCADE,
    FOREIGN KEY (REGION_ID) REFERENCES typeByRegion1(ID) ON DELETE CASCADE);
grant select on recipe_has to public;
```

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```
insert into recipe_has
values(1, To_Date('16 Jun 2020', 'DD Mon YYYY'),4,3,'Taiwanese-Style Three Cup Chicken','r1: first step...',1);
insert into recipe_has
values(2,To_Date('30 Dec 2020','DD Mon YYYY'),2,2,'Sweet and Sour Pork','r2: first step...',2);
insert into recipe_has
values(3,To_Date('17 Jun 2020','DD Mon YYYY'),3,1,'Kung Pao Chicken','r3: first step...',3);
insert into recipe_has
values(4,To_Date('18 Jun 2020','DD Mon YYYY'),4,4,'Cal Smart Steak','r4: first step...',4);
insert into recipe_has
values(5,To_Date('19 Jun 2020','DD Mon YYYY'),5,5,'Beef Udon','r5: first step...',5);
insert into recipe_has
values(6,To_Date('20 Jun 2020','DD Mon YYYY'),6,6,'Poutine','r6: first step...',4);
insert into recipe_has
values(7,To_Date('21 Jun 2020','DD Mon YYYY'),7,7,'Slow Cooker Pot Roast','r7: first step...',3);
insert into recipe_has
values(8,To_Date('22 Jun 2020','DD Mon YYYY'),8,8,'Pasta alla Norma','r8: first step...',2);
insert into recipe_has
values(9,To_Date('23 Jun 2020','DD Mon YYYY'),9,9,'Okonomiyaki','r9: first step...',1);
insert into recipe_has
values(10,To_Date('23 Nov 2021','DD Mon YYYY'),10,10,'Barbecued Pork','r10: first step...',2);
insert into recipe_has
values(11, To_Date('16 Jun 2020','DD Mon YYYY'),10,3,'Taiwanese-Style Three Cup Chicken','r11: first step...',2);
```

```
CREATE TABLE recipeMethod (  
    METHOD_ID INTEGER,  
    RECIPE_ID INTEGER,  
    PRIMARY KEY (METHOD_ID, RECIPE_ID),  
    FOREIGN KEY (METHOD_ID) REFERENCES typeByMethod(ID),  
    FOREIGN KEY (RECIPE_ID) REFERENCES recipe_has(ID) ON DELETE CASCADE);  
grant select on recipeMethod to public;
```

```
insert into recipeMethod  
values(1,1);  
insert into recipeMethod  
values(2,2);  
insert into recipeMethod  
values(3,3);  
insert into recipeMethod  
values(4,4);  
insert into recipeMethod  
values(5,5);  
insert into recipeMethod  
values(2,6);  
insert into recipeMethod  
values(5,6);  
insert into recipeMethod  
values(6,7);  
insert into recipeMethod  
values(5,8);  
insert into recipeMethod  
values(3,9);  
insert into recipeMethod  
values(4,9);  
insert into recipeMethod  
values(7,10);
```

```
CREATE TABLE typeByMethod (  
  ID INTEGER,  
  METHOD CHAR(20),  
  PRIMARY KEY (ID));  
grant select on typeByMethod to public;
```

```
insert into typeByMethod  
values(1,'braise');  
insert into typeByMethod  
values(2,'deep-fried');  
insert into typeByMethod  
values(3,'stir-fried');  
insert into typeByMethod  
values(4,'saute');  
insert into typeByMethod  
values(5,'boil');  
insert into typeByMethod  
values(6,'stew');  
insert into typeByMethod  
values(7,'roast');
```

GUI input:

Search recipes by methods

Method Name (please in lowercase):

Result:

Retrieved data from table:

ID	RECIPE_NAME	STEPS	PICTURE
1	taiwanese-style three cup chicken		

- 6. Join

code reference:

@userInterface.php

```
userInterface.php • managerInterface.php # style.css # style_m.css b...
userInterface.php
...
273 //join contain and recipe_has
274 function handleSearchByIngredient() {
275     global $db_conn;
276     $keyword = $_GET['keyword'];
277     $result = executePlainSQL(["SELECT *
278     FROM CONTAIN c, RECIPE_HAS r
279     WHERE c.INGREDIENT_NAME LIKE '%" . $keyword . "%' and c.recipe_ID = r.ID"]);
280     printSearchResult($result);
281 }
```

Database(before):

```
CREATE TABLE recipe_has (
    ID                INTEGER PRIMARY KEY,
    CREATETIME        DATE,
    USER_ACCOUNT_ID   INTEGER NOT NULL,
    REGION_ID         INTEGER NOT NULL,
    RECIPE_NAME        CHAR(50) NOT NULL,
    DETAILS            CHAR(500),
    DIFFICULTY         SMALLINT,
    FOREIGN KEY (USER_ACCOUNT_ID) REFERENCES user_account(USER_ACCOUNT_ID) ON DELETE CASCADE,
    FOREIGN KEY (REGION_ID) REFERENCES typeByRegion1(ID) ON DELETE CASCADE);
grant select on recipe_has to public;
```

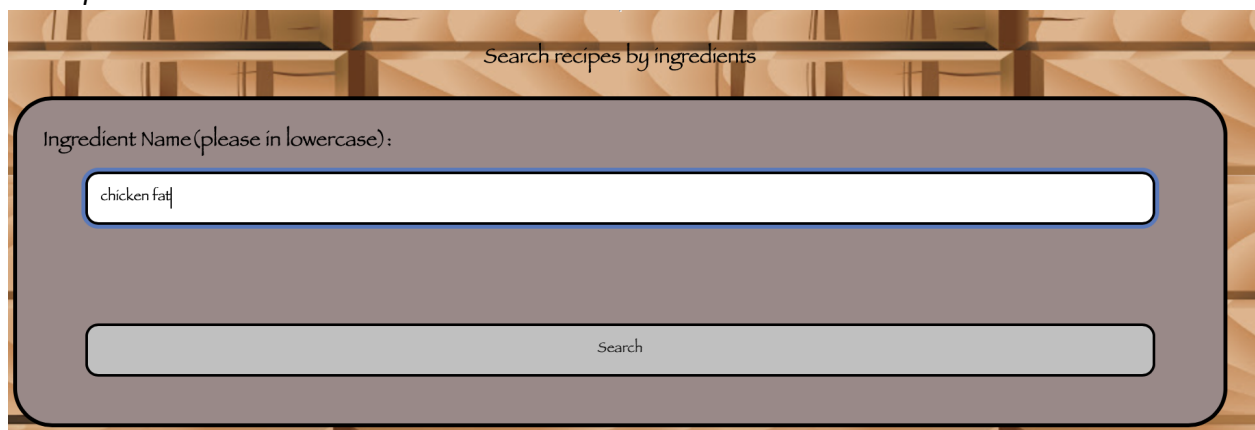
```
insert into recipe_has
values(1, To_Date('16 Jun 2020','DD Mon YYYY'),4,3,'Taiwanese-Style Three Cup Chicken','r1: first step...',1);
insert into recipe_has
values(2,To_Date('30 Dec 2020','DD Mon YYYY'),2,2,'Sweet and Sour Pork','r2: first step...',2);
insert into recipe_has
values(3,To_Date('17 Jun 2020','DD Mon YYYY'),3,1,'Kung Pao Chicken','r3: first step...',3);
insert into recipe_has
values(4,To_Date('18 Jun 2020','DD Mon YYYY'),4,4,'Cal Smart Steak','r4: first step...',4);
insert into recipe_has
values(5,To_Date('19 Jun 2020','DD Mon YYYY'),5,5,'Beef Udon','r5: first step...',5);
insert into recipe_has
values(6,To_Date('20 Jun 2020','DD Mon YYYY'),6,6,'Poutine','r6: first step...',4);
insert into recipe_has
values(7,To_Date('21 Jun 2020','DD Mon YYYY'),7,7,'Slow Cooker Pot Roast','r7: first step...',3);
insert into recipe_has
values(8,To_Date('22 Jun 2020','DD Mon YYYY'),8,8,'Pasta alla Norma','r8: first step...',2);
insert into recipe_has
values(9,To_Date('23 Jun 2020','DD Mon YYYY'),9,9,'Okonomiyaki','r9: first step...',1);
insert into recipe_has
values(10,To_Date('23 Nov 2021','DD Mon YYYY'),10,10,'Barbecued Pork','r10: first step...',2);
insert into recipe_has
values(11, To_Date('16 Jun 2020','DD Mon YYYY'),10,3,'Taiwanese-Style Three Cup Chicken','r11: first step...',2);
```

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```
CREATE TABLE contain (  
    INGREDIENT_NAME      CHAR(50),  
    RECIPE_ID            INTEGER,  
    PRIMARY KEY (INGREDIENT_NAME, RECIPE_ID),  
    FOREIGN KEY (INGREDIENT_NAME) REFERENCES ingredient(INGREDIENT_NAME),  
    FOREIGN KEY (RECIPE_ID) REFERENCES recipe_has(ID) ON DELETE CASCADE);  
grant select on contain to public;
```

```
insert into contain  
values('bacon', 6);  
insert into contain  
values('beef fat', 4);  
insert into contain  
values('butter', 7);  
insert into contain  
values('chicken fat', 1);  
insert into contain  
values('cocoa butter', 2);  
insert into contain  
values('coconut or coconut oil', 2);  
insert into contain  
values('hydrogenated fats and oils', 8);  
insert into contain  
values('lard', 3);  
insert into contain  
values('lard', 2);  
insert into contain  
values('lard', 10);  
insert into contain  
values('palm or palm kernel oil', 5);  
insert into contain  
values('powdered whole milk solids', 9);
```

GUI input:



Search recipes by ingredients

Ingredient Name (please in lowercase):

Search

Result:

Retrieved data from table:

ID	RECIPE_NAME	STEPS	PICTURE
1	taiwanese-style three cup chicken	r2: first step...	

- 7. Aggregation with Group By

Code reference:

@userInterface.php

```
userInterface.php • managerInterface.php # style.css # style_m.css background1.png
userInterface.php
290 // List the number of comments for each recipe name in decreasing order(Aggregation with group by)
291 function handleDisplayPopularRequest(){
292     global $db_conn;
293     $result = executePlainSQL("SELECT count(c.id), r.recipe_name
294     from recipe_has r, comment1 c
295     where r.id = c.recipe_id
296     group by r.recipe_name
297     ORDER BY COUNT(c.id) DESC");
298     printPRResult($result);}
```

Database(before):

```
CREATE TABLE recipe_has (
    ID                INTEGER PRIMARY KEY,
    CREATETIME        DATE,
    USER_ACCOUNT_ID   INTEGER NOT NULL,
    REGION_ID         INTEGER NOT NULL,
    RECIPE_NAME        CHAR(50) NOT NULL,
    DETAILS            CHAR(500),
    DIFFICULTY         SMALLINT,
    FOREIGN KEY (USER_ACCOUNT_ID) REFERENCES user_account(USER_ACCOUNT_ID) ON DELETE CASCADE,
    FOREIGN KEY (REGION_ID) REFERENCES typeByRegion1(ID) ON DELETE CASCADE);
grant select on recipe_has to public;
```


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```
insert into recipe_has
values(1, To_Date('16 Jun 2020','DD Mon YYYY'),4,3,'Taiwanese-Style Three Cup Chicken','r1: first step...',1);
insert into recipe_has
values(2,To_Date('30 Dec 2020','DD Mon YYYY'),2,2,'Sweet and Sour Pork','r2: first step...',2);
insert into recipe_has
values(3,To_Date('17 Jun 2020','DD Mon YYYY'),3,1,'Kung Pao Chicken','r3: first step...',3);
insert into recipe_has
values(4,To_Date('18 Jun 2020','DD Mon YYYY'),4,4,'Cal Smart Steak','r4: first step...',4);
insert into recipe_has
values(5,To_Date('19 Jun 2020','DD Mon YYYY'),5,5,'Beef Udon','r5: first step...',5);
insert into recipe_has
values(6,To_Date('20 Jun 2020','DD Mon YYYY'),6,6,'Poutine','r6: first step...',4);
insert into recipe_has
values(7,To_Date('21 Jun 2020','DD Mon YYYY'),7,7,'Slow Cooker Pot Roast','r7: first step...',3);
insert into recipe_has
values(8,To_Date('22 Jun 2020','DD Mon YYYY'),8,8,'Pasta alla Norma','r8: first step...',2);
insert into recipe_has
values(9,To_Date('23 Jun 2020','DD Mon YYYY'),9,9,'Okonomiyaki','r9: first step...',1);
insert into recipe_has
values(10,To_Date('23 Nov 2021','DD Mon YYYY'),10,10,'Barbecued Pork','r10: first step...',2);
insert into recipe_has
values(11, To_Date('16 Jun 2020','DD Mon YYYY'),10,3,'Taiwanese-Style Three Cup Chicken','r11: first step...',2);
```

```
CREATE TABLE comment1 (
    ID INTEGER,
    CREATE_TIME CHAR(30),
    RECIPE_ID INTEGER,
    USER_ACCOUNT_ID INTEGER,
    PRIMARY KEY (ID),
    FOREIGN KEY (USER_ACCOUNT_ID) REFERENCES user_account(USER_ACCOUNT_ID),
    FOREIGN KEY (RECIPE_ID) REFERENCES recipe_has(ID) ON DELETE CASCADE);
grant select on comment1 to public;
```

```
insert into comment1
values(1, to_date ( '2007-12-20' , 'YYYY-MM-DD' ),11,2);
insert into comment1
values(2, to_date ( '2020-07-16' , 'YYYY-MM-DD' ),2,3);
insert into comment1
values(3, to_date ( '2020-12-30' , 'YYYY-MM-DD' ),1,4);
insert into comment1
values(4, to_date ( '2020-07-17' , 'YYYY-MM-DD' ),3,2);
insert into comment1
values(5, to_date ( '2020-07-28' , 'YYYY-MM-DD' ),4,6);
insert into comment1
values(6, to_date ( '2020-07-29' , 'YYYY-MM-DD' ),5,5);
insert into comment1
values(7, to_date ( '2020-07-23' , 'YYYY-MM-DD' ),6,9);
insert into comment1
values(8, to_date ( '2020-07-25' , 'YYYY-MM-DD' ),7,8);
insert into comment1
values(9, to_date ( '2020-07-05' , 'YYYY-MM-DD' ),3,10);
insert into comment1
values(10, to_date ( '2020-07-01' , 'YYYY-MM-DD' ),1,5);
```

GUI input:



Result:

#COMMENTS	RECIPE_NAME
2	Kung Pao Chicken
2	taiwanese-style three cup chicken
1	Poutine
1	Taiwanese-Style Three Cup Chicken
1	Sweet and Sour Pork
1	Beef Udon
1	Cal Smart Steak
1	Slow Cooker Pot Roast

- 8. Aggregation with Having

Code reference:

In `userInterface.php`

```
userInterface.php • managerInterface.php # style.css # style_m.css background1.png wrapper.php data
userInterface.php
300 printRRResult($result);
307 // LIST THE NUMBER OF COMMENTS FOR EACH RECIPE NAME in decreasing order, ONLY INCLUDES RECIPE NAMES WITH MORE THAN 1 COMMENT.
308 // Aggregation with having
309 function handleDisplayMostPopularRequest(){
310     global $db_conn;
311     $result = executePlainSQL("SELECT count(c.id), r.recipe_name
312 from recipe_has r, comment1 c
313 where r.id = c.recipe_id
314 group by r.recipe_name
315 having count(c.id) > 1
316 ORDER BY COUNT(c.id) DESC");
317 printRRResult($result);}
```

Database(before):

```
CREATE TABLE recipe_has (
    ID                INTEGER PRIMARY KEY,
    CREATETIME        DATE,
    USER_ACCOUNT_ID   INTEGER NOT NULL,
    REGION_ID         INTEGER NOT NULL,
    RECIPE_NAME       CHAR(50) NOT NULL,
    DETAILS           CHAR(500),
    DIFFICULTY        SMALLINT,
    FOREIGN KEY (USER_ACCOUNT_ID) REFERENCES user_account(USER_ACCOUNT_ID) ON DELETE CASCADE,
    FOREIGN KEY (REGION_ID) REFERENCES typeByRegion1(ID) ON DELETE CASCADE);
grant select on recipe_has to public;
```

```
insert into recipe_has
values(1, To_Date('16 Jun 2020','DD Mon YYYY'),4,3,'Taiwanese-Style Three Cup Chicken','r1: first step...',1);
insert into recipe_has
values(2,To_Date('30 Dec 2020','DD Mon YYYY'),2,2,'Sweet and Sour Pork','r2: first step...',2);
insert into recipe_has
values(3,To_Date('17 Jun 2020','DD Mon YYYY'),3,1,'Kung Pao Chicken','r3: first step...',3);
insert into recipe_has
values(4,To_Date('18 Jun 2020','DD Mon YYYY'),4,4,'Cal Smart Steak','r4: first step...',4);
insert into recipe_has
values(5,To_Date('19 Jun 2020','DD Mon YYYY'),5,5,'Beef Udon','r5: first step...',5);
insert into recipe_has
values(6,To_Date('20 Jun 2020','DD Mon YYYY'),6,6,'Poutine','r6: first step...',4);
insert into recipe_has
values(7,To_Date('21 Jun 2020','DD Mon YYYY'),7,7,'Slow Cooker Pot Roast','r7: first step...',3);
insert into recipe_has
values(8,To_Date('22 Jun 2020','DD Mon YYYY'),8,8,'Pasta alla Norma','r8: first step...',2);
insert into recipe_has
values(9,To_Date('23 Jun 2020','DD Mon YYYY'),9,9,'Okonomiyaki','r9: first step...',1);
insert into recipe_has
values(10,To_Date('23 Nov 2021','DD Mon YYYY'),10,10,'Barbecued Pork','r10: first step...',2);
insert into recipe_has
values(11, To_Date('16 Jun 2020','DD Mon YYYY'),10,3,'Taiwanese-Style Three Cup Chicken','r11: first step...',2);
```

```
CREATE TABLE comment1 (
    ID INTEGER,
    CREATE_TIME CHAR(30),
    RECIPE_ID INTEGER,
    USER_ACCOUNT_ID INTEGER,
    PRIMARY KEY (ID),
    FOREIGN KEY (USER_ACCOUNT_ID) REFERENCES user_account(USER_ACCOUNT_ID),
    FOREIGN KEY (RECIPE_ID) REFERENCES recipe_has(ID) ON DELETE CASCADE);
grant select on comment1 to public;
```

```
insert into comment1
values(1, to_date ( '2007-12-20' , 'YYYY-MM-DD' ),11,2);
insert into comment1
values(2, to_date ( '2020-07-16' , 'YYYY-MM-DD' ),2,3);
insert into comment1
values(3, to_date ( '2020-12-30' , 'YYYY-MM-DD' ),1,4);
insert into comment1
values(4, to_date ( '2020-07-17' , 'YYYY-MM-DD' ),3,2);
insert into comment1
values(5, to_date ( '2020-07-28' , 'YYYY-MM-DD' ),4,6);
insert into comment1
values(6, to_date ( '2020-07-29' , 'YYYY-MM-DD' ),5,5);
insert into comment1
values(7, to_date ( '2020-07-23' , 'YYYY-MM-DD' ),6,9);
insert into comment1
values(8, to_date ( '2020-07-25' , 'YYYY-MM-DD' ),7,8);
insert into comment1
values(9, to_date ( '2020-07-05' , 'YYYY-MM-DD' ),3,10);
insert into comment1
values(10, to_date ( '2020-07-01' , 'YYYY-MM-DD' ),1,5);
```

GUI input:



Result:

#COMMENTS	RECIPE_NAME
2	Kung Pao Chicken
2	taiwanese-style three cup chicken

- 9. Nested Aggregation with Group By

Code reference:

In `userInterface.php`

```
userInterface.php • managerInterface.php # style.css # style_m.css background1.png wrapper.php dat
userInterface.php
299 // return the recipe name and steps with highest average rating(nested Aggregation with group by)
300 function handleDisplayRequest(){
301     global $db_conn;
302     $result = executePlainSQL("SELECT r.recipe_name,r.details
303     from recipe_has r, rating ra3
304     where ra3.recipe_id = r.id and
305     ra3.score = (SELECT MAX(AVG(score)) FROM RATING ra, RECIPE_HAS r where r.id = ra.recipe_id group by r.recipe_name)");
306     printHRRResult($result);}
```

Database(before):

```
CREATE TABLE recipe_has (
  ID INTEGER PRIMARY KEY,
  CREATETIME DATE,
  USER_ACCOUNT_ID INTEGER NOT NULL,
  REGION_ID INTEGER NOT NULL,
  RECIPE_NAME CHAR(50) NOT NULL,
  DETAILS CHAR(500),
  DIFFICULTY SMALLINT,
  FOREIGN KEY (USER_ACCOUNT_ID) REFERENCES user_account(USER_ACCOUNT_ID) ON DELETE CASCADE,
  FOREIGN KEY (REGION_ID) REFERENCES typeByRegion1(ID) ON DELETE CASCADE);
grant select on recipe_has to public;
```

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```
insert into recipe_has
values(1, To_Date('16 Jun 2020', 'DD Mon YYYY'), 4, 3, 'Taiwanese-Style Three Cup Chicken', 'r1: first step...', 1);
insert into recipe_has
values(2, To_Date('30 Dec 2020', 'DD Mon YYYY'), 2, 2, 'Sweet and Sour Pork', 'r2: first step...', 2);
insert into recipe_has
values(3, To_Date('17 Jun 2020', 'DD Mon YYYY'), 3, 1, 'Kung Pao Chicken', 'r3: first step...', 3);
insert into recipe_has
values(4, To_Date('18 Jun 2020', 'DD Mon YYYY'), 4, 4, 'Cal Smart Steak', 'r4: first step...', 4);
insert into recipe_has
values(5, To_Date('19 Jun 2020', 'DD Mon YYYY'), 5, 5, 'Beef Udon', 'r5: first step...', 5);
insert into recipe_has
values(6, To_Date('20 Jun 2020', 'DD Mon YYYY'), 6, 6, 'Poutine', 'r6: first step...', 4);
insert into recipe_has
values(7, To_Date('21 Jun 2020', 'DD Mon YYYY'), 7, 7, 'Slow Cooker Pot Roast', 'r7: first step...', 3);
insert into recipe_has
values(8, To_Date('22 Jun 2020', 'DD Mon YYYY'), 8, 8, 'Pasta alla Norma', 'r8: first step...', 2);
insert into recipe_has
values(9, To_Date('23 Jun 2020', 'DD Mon YYYY'), 9, 9, 'Okonomiyaki', 'r9: first step...', 1);
insert into recipe_has
values(10, To_Date('23 Nov 2021', 'DD Mon YYYY'), 10, 10, 'Barbecued Pork', 'r10: first step...', 2);
insert into recipe_has
values(11, To_Date('16 Jun 2020', 'DD Mon YYYY'), 10, 3, 'Taiwanese-Style Three Cup Chicken', 'r11: first step...', 2);
```

```
CREATE TABLE rating (
    USER_ACCOUNT_ID      INTEGER,
    RECIPE_ID             INTEGER,
    SCORE                 INTEGER,
    PRIMARY KEY (USER_ACCOUNT_ID, RECIPE_ID),
    FOREIGN KEY (USER_ACCOUNT_ID) REFERENCES user_account(USER_ACCOUNT_ID),
    FOREIGN KEY (RECIPE_ID) REFERENCES recipe_has(ID) ON DELETE CASCADE);

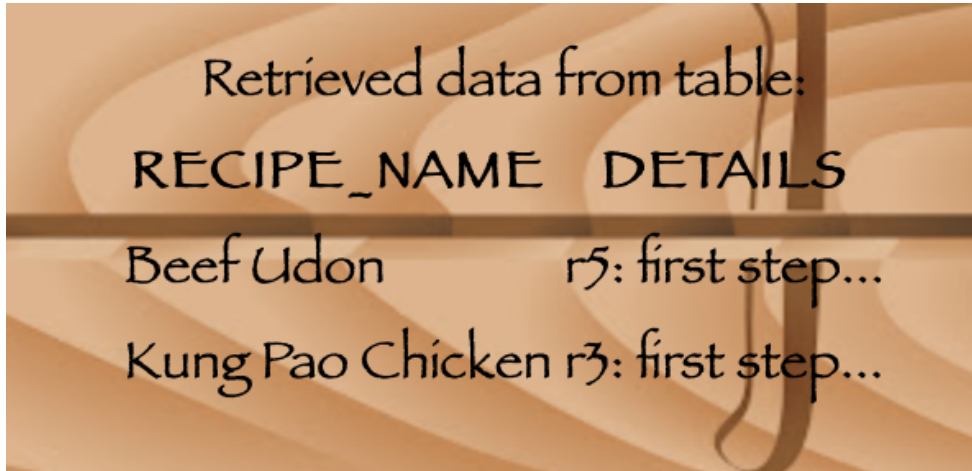
grant select on rating to public;
```

```
insert into rating
values(7, 2, 3);
insert into rating
values(7, 3, 4);
insert into rating
values(2, 4, 3);
insert into rating
values(7, 5, 5);
insert into rating
values(7, 8, 3);
insert into rating
values(10, 2, 3);
insert into rating
values(4, 1, 1);
insert into rating
values(5, 3, 5);
insert into rating
values(2, 9, 1);
insert into rating
values(8, 7, 2);
```

GUI input:



Result:



- 10. Division

Code reference:

In `userInterface.php`

```
userInterface.php • managerInterface.php # style.css # style_m.css ba
userInterface.php
318 //list the recipe names without selected gradient(Division)
319 function handleDivisionRecipeRequest(){
320     global $db_conn;
321     $keyword = $_POST['ans'];
322     echo 'There are the recipes without ', $keyword;
323     $result = executePlainSQL("SELECT r1.recipe_name
324 from recipe_has r1
325 minus
326 select r2.recipe_name
327 from recipe_has r2, contain c
328 where r2.id = c.recipe_id and c.ingredient_name LIKE '%" . $keyword . "%'");
329 printDRResult($result);}
330
```

Database(before):

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```
CREATE TABLE recipe_has (  
    ID                INTEGER PRIMARY KEY,  
    CREATETIME        DATE,  
    USER_ACCOUNT_ID   INTEGER NOT NULL,  
    REGION_ID         INTEGER NOT NULL,  
    RECIPE_NAME        CHAR(50) NOT NULL,  
    DETAILS            CHAR(500),  
    DIFFICULTY         SMALLINT,  
    FOREIGN KEY (USER_ACCOUNT_ID) REFERENCES user_account(USER_ACCOUNT_ID) ON DELETE CASCADE,  
    FOREIGN KEY (REGION_ID) REFERENCES typeByRegion1(ID) ON DELETE CASCADE);  
grant select on recipe_has to public;
```

```
insert into recipe_has  
values(1, To_Date('16 Jun 2020', 'DD Mon YYYY'), 4, 3, 'Taiwanese-Style Three Cup Chicken', 'r1: first step...', 1);  
insert into recipe_has  
values(2, To_Date('30 Dec 2020', 'DD Mon YYYY'), 2, 2, 'Sweet and Sour Pork', 'r2: first step...', 2);  
insert into recipe_has  
values(3, To_Date('17 Jun 2020', 'DD Mon YYYY'), 3, 1, 'Kung Pao Chicken', 'r3: first step...', 3);  
insert into recipe_has  
values(4, To_Date('18 Jun 2020', 'DD Mon YYYY'), 4, 4, 'Cal Smart Steak', 'r4: first step...', 4);  
insert into recipe_has  
values(5, To_Date('19 Jun 2020', 'DD Mon YYYY'), 5, 5, 'Beef Udon', 'r5: first step...', 5);  
insert into recipe_has  
values(6, To_Date('20 Jun 2020', 'DD Mon YYYY'), 6, 6, 'Poutine', 'r6: first step...', 4);  
insert into recipe_has  
values(7, To_Date('21 Jun 2020', 'DD Mon YYYY'), 7, 7, 'Slow Cooker Pot Roast', 'r7: first step...', 3);  
insert into recipe_has  
values(8, To_Date('22 Jun 2020', 'DD Mon YYYY'), 8, 8, 'Pasta alla Norma', 'r8: first step...', 2);  
insert into recipe_has  
values(9, To_Date('23 Jun 2020', 'DD Mon YYYY'), 9, 9, 'Okonomiyaki', 'r9: first step...', 1);  
insert into recipe_has  
values(10, To_Date('23 Nov 2021', 'DD Mon YYYY'), 10, 10, 'Barbecued Pork', 'r10: first step...', 2);  
insert into recipe_has  
values(11, To_Date('16 Jun 2020', 'DD Mon YYYY'), 10, 3, 'Taiwanese-Style Three Cup Chicken', 'r11: first step...', 2);
```

```
CREATE TABLE contain (  
    INGREDIENT_NAME   CHAR(50),  
    RECIPE_ID          INTEGER,  
    PRIMARY KEY (INGREDIENT_NAME, RECIPE_ID),  
    FOREIGN KEY (INGREDIENT_NAME) REFERENCES ingredient(INGREDIENT_NAME),  
    FOREIGN KEY (RECIPE_ID) REFERENCES recipe_has(ID) ON DELETE CASCADE);  
grant select on contain to public;
```



```
insert into contain
values('bacon', 6);
insert into contain
values('beef fat', 4);
insert into contain
values('butter', 7);
insert into contain
values('chicken fat', 1);
insert into contain
values('cocoa butter', 2);
insert into contain
values('coconut or coconut oil', 2);
insert into contain
values('hydrogenated fats and oils', 8);
insert into contain
values('lard', 3);
insert into contain
values('lard', 2);
insert into contain
values('lard', 10);
insert into contain
values('palm or palm kernel oil', 5);
insert into contain
values('powdered whole milk solids', 9);
```

GUI input:



avoid certain food?

Get your wanted diet!

☒ I don't want food containing milk

☐ I don't want food containing fat

☐ I don't want food containing bacon

Submit

Result:



*Thank you for viewing this long long document.
Wish you have a good day.*